

FRANK: Flexible Recurrent Active-memory Neural Kernel

A Brain-Inspired Modular Architecture for Extreme Length Generalization

Izen Thornton

Independent Researcher

izen@cixate.com

Abstract

On a state-transition task, 3 of 10 seeds of a modular recurrent architecture maintain perfect accuracy (zero errors) at $100,000\times$ training length (2,000,000 tokens). No other tested architecture achieves this; the best GRU seed reaches 85.5% at the same scale. The outcome is strongly seed-dependent: two seeds hold at $\sim 90\%$, two degrade to 55–62%, three fail at chance. Same architecture, same training, same data; initialization alone determines whether the model finds a perfectly generalizing solution.

The architecture is FRANK (Flexible Recurrent Active-memory Neural Kernel), built through a three-step progression (Modular $\rightarrow$ Modular+Memory $\rightarrow$ FRANK), each stage adding one neuroscience-inspired component. Each module of FRANK is architecturally different (recurrence, memory, feedforward MLP), allowing gradient descent alone to determine what module to use depending on the task. A veto layer blends brain and reflex outputs; a content-addressable memory injects stored values into the recurrent modules; an anomaly detector monitors for distributional shift. I evaluate FRANK and six baselines across four algorithmic sequence tasks using ten random seeds. A lateral connection ablation shows inter-module communication changes which seeds succeed rather than uniformly helping. A negative result with structured multi-timescale memory shows that added complexity hurts; architectural minimalism acts as implicit regularization.

1 Introduction

Length generalization is the ability of a neural network to extrapolate learned algorithms to sequences far longer than those seen during training. It remains an open problem. Transformers fail on algorithmic tasks [2, 3]. Standard recurrent networks like LSTMs and GRUs suffer from state instability over long sequences [4, 5]. The Neural GPU achieves around $100\times$ generalization on binary addition, but it requires curriculum learning and has been shown to be fragile [1, 6]. Memory-augmented architectures like Neural Turing Machines [7] address state capacity but add complex addressing mechanisms.

Neuroscience takes a different approach. The brain handles flexible computation through modular specialization: the hippocampus for memory [8], the basal ganglia for action selection [9], the amygdala for novelty detection [10], the brainstem for fast reactive processing. Multiple researchers have argued that neuroscience should inform AI design more directly [11], but few modern architectures actually map brain regions to differentiable modules.

I built FRANK to test whether that mapping produces useful properties. The brain-region labels (cortex, hippocampus, brainstem, amygdala, basal ganglia) are loose functional analogies, not mechanistic claims; the components were inspired by those systems but do not simulate them. FRANK is a modular recurrent architecture developed in three stages (Modular, Modular+Memory, FRANK), each adding a neuroscience-inspired component. It trains end-to-end with standard backpropagation. The main result is that on a state-transition task, 3 of 10 seeds achieve perfect generalization at $100,000\times$ training length, while 3 other seeds fail entirely. No other tested architecture produces perfect generalization at this scale. I also report a lateral connection ablation and a negative result with multi-timescale memory.

2 Related Work

2.1 Length Generalization

Anil et al. [3] showed that scaling data, model size, or compute does not fix transformer length generalization. Zhou et al. [12] got transformers to $\sim 2.5\times$ on addition with careful data formatting, but noted the results are “highly sensitive to initialization.” Kaiser and Sutskever [1] reached $\sim 100\times$ with Neural GPUs; Price et al. [6] later showed the approach is fragile. Delétang et al. [13] tested many architectures across the Chomsky hierarchy and found LSTMs handle regular and counter languages, but did not report extreme extrapolation ratios. Looped Transformers [14] generalize well through adaptive computation but require architectural changes.

Zhou et al.’s observation about initialization sensitivity is relevant here. My results extend it to a different regime: not marginal sensitivity, but a discrete split between perfect, partial, and complete failure at $100,000\times$.

2.2 Brain-Inspired Architectures

Hassabis et al. [11] made the case for neuroscience-informed AI. Complementary Learning Systems theory [8] motivates separating fast and slow learning. Computational models of the basal ganglia [9] and amygdala [10] directly motivated FRANK’s veto layer and anomaly detector. Linsley et al. [15] showed lateral recurrent connections enable capabilities that feedforward processing cannot achieve alone. Spoerer et al. [16] showed lateral connections improve recognition of degraded inputs.

3 Architecture

FRANK is the last of three architectures I built, where each one was motivated by the limitations of the previous. All three (Modular, Modular+Memory, FRANK) are original to this work. They share core components (tau-gated modules, lateral connections) but were implemented independently; hidden dimensions and output heads differ, so this is a development trajectory rather than a strict component-removal ablation.

3.1 Recurrent Modules

All three architectures use the same building block: tau-gated recurrent modules. These are simpler than GRUs; no gates, just a tau-damped recurrence:

$$h_t = \left(1 - \frac{1}{\tau}\right) h_{t-1} + \frac{1}{\tau} \tanh(W_x x_t + W_h h_{t-1}) \quad (1)$$

τ is learnable (stored as $\log \tau$, exponentiated, clamped to $[1, 100]$). High τ means slow change; low τ means fast response. The four modules are initialized at $\tau_{\text{init}} \in \{5, 15, 30, 50\}$, creating a multi-timescale hierarchy that adapts during training. Lateral connections are applied after module updates: $h \leftarrow h + 0.1 \cdot W_{\text{lat}} \cdot h_{\text{all}}$. The 0.1 scaling is hardcoded to prevent laterals from dominating.

3.2 Architectural Progression

Table 1: Architectural progression. Hidden dimensions and output heads differ between models; this is a development trajectory, not strict ablation.

	Modules	Laterals	Memory	Reflex	Anomaly	Params
Modular	4×209	✓				$\sim 502\text{K}$
Modular+Mem	4×140	✓	✓			$\sim 512\text{K}$
FRANK	4×180	✓	✓	✓	✓	$\sim 507\text{K}$

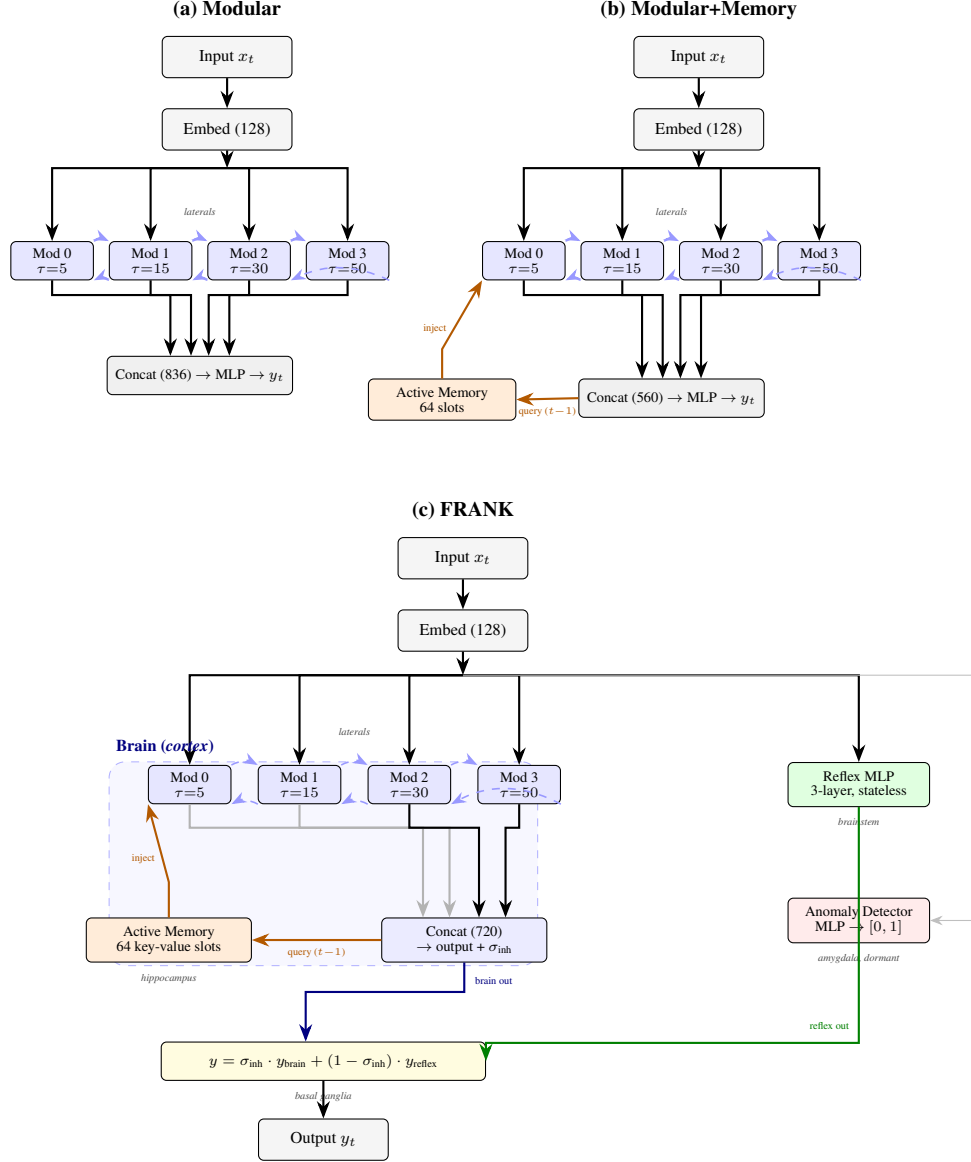


Figure 1: Architectural progression. (a) Modular: four tau-gated recurrent modules with lateral connections, output via 2-layer MLP. (b) Modular+Memory: adds content-addressable active memory (64 slots) injected into module 0. (c) FRANK: adds reflex pathway, anomaly detector, and veto mechanism. The inhibit signal σ_{inh} blends brain and reflex outputs. All components active every timestep; no routing.

Modular ($\sim 502\text{K}$ params): Four tau-gated modules with laterals. Hidden states are concatenated and passed through a 2-layer MLP (ReLU, dropout) to produce output. No memory, reflex, or anomaly detection.

Modular+Memory ($\sim 512\text{K}$ params): Same structure, plus a content-addressable active memory with 64 learned key-value slots. Each timestep, the *previous* timestep’s combined hidden state generates a query through a relevance network; softmax attention over stored keys retrieves a value that gets injected into module 0. Module hidden dimensions shrink from 209 to 140 to keep the total parameter count comparable. Output still uses a 2-layer MLP.

FRANK ($\sim 507\text{K}$ params): Adds three components. A feedforward reflex pathway (3-layer MLP, no recurrence) provides fast stateless responses. An anomaly detector (small MLP outputting a scalar score) monitors for distributional shift; it has no direct connection to the output blend and trains only through gradient flow, remaining dormant on the clean tasks tested here. A veto mechanism blends brain and reflex: the brain produces a sigmoid inhibit signal (bias initialized at -1.0 , so $\sigma(-1) \approx 0.27$), meaning FRANK starts $\sim 73\%$ reflex-dominated and *learns* to trust the brain. The output is: $y = \sigma_{\text{inhibit}} \cdot y_{\text{brain}} + (1 - \sigma_{\text{inhibit}}) \cdot y_{\text{reflex}}$. Module hidden dimensions are 180; the brain uses a single linear projection rather than the 2-layer MLP of earlier models. All components are active every timestep. No routing, no task signal, no component-specific loss.

Every other modular architecture I am aware of uses routing to decide which modules process which inputs. RIMs [17] use attention-based input selection. MoE architectures [18] use gating networks. FRANK does not. All components see all inputs all the time. Lateral connections are applied after individual module updates (not during) and I hypothesize they provide error correction that prevents state drift over long sequences [15, 16].

4 Experimental Setup

4.1 Baselines

I compare the three original architectures against four external baselines at comparable parameter counts: Transformer [2] ($\sim 481\text{K}$), GRU [5] ($\sim 500\text{K}$), RIMs [17] ($\sim 500\text{K}$), and FRANK-NoLat ($\sim 507\text{K}$), which is FRANK with lateral connections replaced by an identity operation.

4.2 Tasks

Chain: State-transition lookups. Step-local; only the current state and current input matter at each step, so perfect generalization is theoretically achievable. **Copy:** Reproduce a token sequence in order. **Recall:** Remember specific tokens through intervening noise. **Sum:** Running cumulative sum, after Trask et al. [19].

4.3 Protocol

All models train on sequences of length 5–20 with a vocabulary of 10 tokens. I evaluate at standard generalization scales ($2\times$, $3\times$, $5\times$, $10\times$) and, for the chain task, at extreme scales up to $100,000\times$ ($2,000,000$ tokens). Ten random seeds: 42, 123, 456, 789, 1337, 2024, 3141, 4242, 5555, 6789. Optimizer is AdamW at 3×10^{-4} . FRANK trains for 200 epochs; baselines train for 100 with early stopping. All experiments ran on NVIDIA RTX 3080. Code, model definitions, training configurations, and raw training logs are available at <https://github.com/izenthornton/frank-papers>

5 Results

5.1 Extreme Length Generalization: Chain Task

Table 2 shows chain task accuracy at $100,000\times$ for all ten seeds.

Three FRANK seeds hit 100.0% accuracy across 2 million tokens. Two hold around 90%. Two partially degrade to 55–62%. Three fail at chance ($\sim 10\%$). The GRU never gets above 85.5%; all its seeds degrade monotonically.

Table 2: Chain task accuracy at $100,000\times$ training length (2,000,000 tokens) per seed.

Seed	FRANK	GRU	FRANK-NoLat
1337	100.0%	14.5%	10.1%
2024	100.0%	15.8%	100.0%
3141	100.0%	27.3%	9.9%
42	91.8%	16.4%	100.0%
789	90.1%	36.1%	89.9%
123	55.3%	28.9%	90.4%
456	62.4%	85.5%	10.2%
4242	10.1%	20.0%	9.9%
5555	9.9%	16.1%	10.1%
6789	10.0%	67.9%	18.2%
Mean	63.0%	32.8%	44.9%

Same architecture, same data, same hyperparameters. Initialization is the only variable. The seeds that do generalize exceed the Neural GPU’s $\sim 100\times$ [1] by up to three orders of magnitude.

Table 3: Mean chain accuracy across scales (10 seeds). FRANK overtakes GRU beyond $1,000\times$.

Scale	Tokens	FRANK	GRU	FRANK-NL	Seeds $>90\%$
$10\times$	200	88.0%	100.0%	81.6%	7/10
$100\times$	2K	74.7%	100.0%	63.1%	7/10
$1,000\times$	20K	72.8%	93.5%	57.1%	6/10
$10,000\times$	200K	70.7%	68.1%	51.8%	5/10
$30,000\times$	600K	67.9%	48.0%	47.5%	4/10
$100,000\times$	2M	63.0%	32.8%	44.9%	3/10

At moderate scales, GRU’s mean is higher because FRANK’s three failed seeds drag the average down. But GRU degrades monotonically while FRANK’s successful seeds hold. The crossover happens between $1,000\times$ and $10,000\times$. After that, FRANK pulls ahead.

5.2 Standard Generalization Across Tasks

Table 4 shows $2\times$ – $10\times$ generalization (10-seed means).

No architecture wins everything. GRU dominates chain at moderate scales and recall (large monolithic hidden states are good at passive storage). Modular+Memory leads on sum. FRANK’s advantage is specifically extreme-scale chain generalization; it is the only architecture where any seed achieves perfect accuracy past $10,000\times$. Copy results are omitted from the main tables because every architecture collapses to chance ($\sim 10\%$) by $3\times$; the full data is in the training logs.

5.3 Lateral Connection Ablation

Laterals do not uniformly help. Seed 1337 generalizes perfectly with laterals but fails without them. Seed 42 does the opposite: 100% without laterals, 91.8% with. Seed 2024 hits 100.0% either way, finding the generalizing basin regardless. On standard generalization (2 – $10\times$), FRANK with laterals consistently beats the ablated variant. But at extreme scales, the effect is seed-specific. Laterals change which initializations lead to generalizing solutions rather than making all of them better.

Table 4: Standard generalization (10-seed mean token accuracy). Top-2 per column in bold.

Model	Chain				Sum			
	2×	3×	5×	10×	2×	3×	5×	10×
FRANK	99.8	99.1	95.8	87.5	89.1	75.7	57.3	41.0
GRU	100.0	100.0	100.0	100.0	94.5	87.0	71.8	49.3
RIMs	98.5	95.6	88.2	77.3	89.4	78.6	61.9	41.7
Modular	94.8	75.1	48.3	28.4	96.4	84.6	63.8	44.7
Mod+Mem	90.6	73.0	49.2	29.5	97.2	86.7	67.4	47.4
Transformer	18.5	14.4	12.0	10.7	7.8	4.9	2.8	1.4

Model	Recall			
	2×	3×	5×	10×
Transformer	100.0	100.0	100.0	100.0
GRU	100.0	100.0	100.0	96.1
Modular	98.7	87.3	81.2	90.0
FRANK	96.9	77.6	61.8	54.8
Mod+Mem	95.4	67.6	58.0	68.4
RIMs	89.6	66.4	45.5	13.3

6 Negative Result: RAM-FRANK

I tested a variant (RAM-FRANK) that replaces the single Active Memory with 16 slots, each with a learnable write-gate bias initialized from 0.0 to 2.0. It matched in-distribution accuracy at 99.8%. Generalization collapsed from 72.8% to 24.1% at 2× on copy. The slots did differentiate (learned biases ranged from -0.83 to 1.32), but the model learned brittle position-specific storage.

Simpler memory tends to outperform structured memory. Fast weight programmers [20] beat LSTMs on certain tasks. The Gated Transformer-XL beats MERLIN’s explicit external memory [21]. The pattern holds here: FRANK’s single monolithic memory forces generalizable compression. Adding slots lets the model overfit to training length. Architectural minimalism acts as implicit regularization.

7 Discussion

7.1 Seed-Dependent Generalization

The most interesting result is the discrete clustering of outcomes. Prior work [12] noted initialization sensitivity, but what I see is different: outcomes cluster at 100%, $\sim 55\text{--}92\%$, and $\sim 10\%$ with nothing in between. The loss landscape appears to contain separate basins that correspond to different learned algorithms. Some basins produce a solution that perfectly implements the state-transition computation; others produce approximations that break at scale.

The three failed seeds never learn the algorithmic structure at all; they are near-chance even at 10×. The two partial seeds achieve perfect accuracy through 1,000× before degrading, which looks like a nearly-correct algorithm with slow state drift. What distinguishes these basins at initialization is an open question.

7.2 Lateral Connections and Error Correction

I believe laterals provide error correction by letting each module’s hidden state get checked against its neighbors, preventing drift. Linsley et al. [15] and Spoerer et al. [16] support this interpretation in other contexts. The ablation is consistent with this idea, but laterals also change *which* seeds generalize rather than uniformly improving all of them.

They reshape the optimization surface, not just stabilize it.

7.3 Limitations

Extreme chain generalization is seed-dependent: only 3 of 10 seeds are perfect. All four tasks are synthetic; I have not tested on natural language or continuous control. The anomaly detector is dormant across all tasks. I do not yet know what initialization properties distinguish generalizing seeds from non-generalizing ones.

8 Conclusion

FRANK achieves perfect generalization at $100,000\times$ training length in 3 of 10 seeds, with $> 90\%$ accuracy in 5 of 10. No other tested architecture produces perfect generalization at this scale. The seed-dependency of the outcome is itself a finding: same architecture, same training, and initialization alone determines whether the model discovers a perfectly generalizing algorithm. Lateral connections change which seeds succeed. Adding memory complexity hurts. Simplicity helps.

Part II analyzes emergent specialization via lesion studies.

References

- [1] Kaiser, L. & Sutskever, I. Neural GPUs Learn Algorithms. *ICLR*, 2016.
- [2] Vaswani, A., Shazeer, N., Parmar, N., et al. Attention Is All You Need. *NeurIPS*, 2017.
- [3] Anil, C., Wu, Y., Andreassen, A., et al. Exploring Length Generalization in Large Language Models. *NeurIPS*, 2022.
- [4] Hochreiter, S. & Schmidhuber, J. Long Short-Term Memory. *Neural Computation*, 9(8):1735–1780, 1997.
- [5] Cho, K., van Merriënboer, B., Gulcehre, C., et al. On the Properties of Neural Machine Translation: Encoder–Decoder Approaches. *SSST-8*, 2014.
- [6] Price, E., Zaremba, W. & Sutskever, I. Extensions and Limitations of the Neural GPU. *ICLR Workshop*, 2017.
- [7] Graves, A., Wayne, G. & Danihelka, I. Neural Turing Machines. *arXiv:1410.5401*, 2014.
- [8] McClelland, J.L., McNaughton, B.L. & O’Reilly, R.C. Why There Are Complementary Learning Systems. *Psychological Review*, 102(3):419–457, 1995.
- [9] Frank, M.J., Loughry, B. & O’Reilly, R.C. Interactions Between Frontal Cortex and Basal Ganglia in Working Memory. *CABN*, 1(2):137–160, 2001.
- [10] Blackford, J.U., Buckholz, J.W., Avery, S.N. & Zald, D.H. A Unique Role for the Human Amygdala in Novelty Detection. *NeuroImage*, 50(3):1188–1193, 2010.
- [11] Hassabis, D., Kumaran, D., Summerfield, C. & Botvinick, M. Neuroscience-Inspired Artificial Intelligence. *Neuron*, 95(2):245–258, 2017.
- [12] Zhou, H., Bradley, A., Littwin, E., et al. Transformers Can Achieve Length Generalization But Not Robustly. *arXiv:2402.09371*, 2024.
- [13] Delétang, L., Ruoss, A., Grau-Moya, J., et al. Neural Networks and the Chomsky Hierarchy. *ICLR*, 2023.
- [14] Fan, Y., Du, Y., Ramchandran, K. & Lee, K. Looped Transformers for Length Generalization. *ICLR*, 2025.
- [15] Linsley, D., Kim, J., Veerabadran, V., et al. Learning Long-Range Spatial Dependencies with Horizontal Gated Recurrent Units. *NeurIPS*, 2018.

- [16] Spoerer, C.J., McClure, P. & Kriegeskorte, N. Recurrent Convolutional Neural Networks: A Better Model of Biological Object Recognition. *Frontiers in Psychology*, 8:1551, 2017.
- [17] Goyal, A., Lamb, A., Hoffmann, J., et al. Recurrent Independent Mechanisms. *ICLR*, 2021.
- [18] Jacobs, R.A., Jordan, M.I., Nowlan, S.J. & Hinton, G.E. Adaptive Mixtures of Local Experts. *Neural Computation*, 3(1):79–87, 1991.
- [19] Trask, A., Hill, F., Reed, S., et al. Neural Arithmetic Logic Units. *NeurIPS*, 2018.
- [20] Schmidhuber, J. Learning to Control Fast-Weight Memories. *Neural Computation*, 4(1):131–139, 1992.
- [21] Parisotto, E., Song, F., Rae, J., et al. Stabilizing Transformers for Reinforcement Learning. *ICML*, 2020.